# DESCENT ALONG A FAITHFULLY FLAT RING MAP OF FINITE TYPE, FINITE PRESENTATION, FORMAL UNRAMIFIEDNESS, UNRAMIFIEDNESS AND ETALENESS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof cites [EGA4] twice, once as an alternative to the Algebra lemma used in part (c) and once as an alternative in part (e). Both citations lie in the delivered proof text and are reproduced verbatim; they are redundant with the same-chapter-external Algebra lemmas that the author names first, so no mathematical content depends on them.

*Editorial note.* The counted spans carry 4 navigation stubs; no same-chapter context span is required and none was added.

*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- Descent of module properties along a faithfully flat ring map. Same chapter, Theorem theorem-descent, invoked at proof lines 1710-1711. Turns the vanishing of a base changed module into the vanishing of the module itself, which is how parts (c) and (e) conclude.
- Finite generation of the kernel of a map of polynomial algebras can be checked after faithfully flat base change. Algebra, Lemma algebra-lemma-finite-presentation-independent, invoked at proof lines 1694-1696. Supplies the finite set of elements from which the finitely presented approximation A' of A is built in part (b).
- The differential criterion for formal unramifiedness and the flat, unramified, finitely presented criterion for etaleness. Algebra, Lemma algebra-lemma-characterize-formally-unramified at proof lines 1706-1708, Algebra, Lemma algebra-lemma-formally-unramified-unramified at proof lines 1714-1717, Algebra, Lemma algebra-lemma-etale-flat-unramified-finite-presentation at proof lines 1720-1722. Convert the properties being descended into conditions on the module of relative differentials, on finite type, and on flatness.

*Source reference.* Original lemma 00UO: sources/stacks-chow/tags.txt, source lines 1118–1118. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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*Date*: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 08WD: sources/stacks-chow/tags.txt, source lines 11266–11266. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 00R2: sources/stacks-chow/tags.txt, source lines 988–988. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 00UU: sources/stacks-chow/tags.txt, source lines 1124–1124. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original theorem 08XA: sources/stacks-descent/pinned-original-source.tex, source lines 1507–1507. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Theorem 0.1.** *If $A \otimes_R S$ has one of the following properties as an $S$-algebra*

(a) *of finite type;*
(b) *of finite presentation;*
(c) *formally unramified;*
(d) *unramified;*
(e) *étale;*

*then so does $A$ as an $R$-algebra (and of course conversely).*

*Proof.* To prove (a), choose a finite set $\{x_i\}$ of generators of $A \otimes_R S$ over $S$. Write each $x_i$ as $\sum_j y_{ij} \otimes s_{ij}$ with $y_{ij} \in A$ and $s_{ij} \in S$. Let $F$ be the polynomial $R$-algebra on variables $e_{ij}$ and let $F \to M$ be the $R$-algebra map sending $e_{ij}$ to $y_{ij}$. Then $F \otimes_R S \to A \otimes_R S$ is surjective, so $\mathrm{Coker}(F \to A) \otimes_R S$ is zero and hence $\mathrm{Coker}(F \to A)$ is zero. This proves (a).

To see (b) assume $A \otimes_R S$ is a finitely presented $S$-algebra. Then $A$ is finite type over $R$ by (a). Choose a surjection $R[x_1, \ldots, x_n] \to A$ with kernel $I$. Then $I \otimes_R S \to S[x_1, \ldots, x_n] \to A \otimes_R S \to 0$ is exact. By Algebra, Lemma 00R2 the kernel of $S[x_1, \ldots, x_n] \to A \otimes_R S$ is a finitely generated ideal. Thus we can find finitely many elements $y_1, \ldots, y_t \in I$ such that the images of $y_i \otimes 1$ in $S[x_1, \ldots, x_n]$ generate the kernel of $S[x_1, \ldots, x_n] \to A \otimes_R S$. Let $I' \subset I$ be the ideal generated by $y_1, \ldots, y_t$. Then $A' = R[x_1, \ldots, x_n]/I'$ is a finitely presented $R$-algebra with a morphism $A' \to A$ such that $A' \otimes_R S \to A \otimes_R S$ is an isomorphism. Thus $A' \cong A$ as desired.

To prove (c), recall that $A$ is formally unramified over $R$ if and only if the module of relative differentials $\Omega_{A/R}$ vanishes, see Algebra, Lemma 00UO or [EGA4, Proposition 17.2.1]. Since $\Omega_{(A \otimes_R S)/S} = \Omega_{A/R} \otimes_R S$, the vanishing descends by Theorem 08XA.

To deduce (d) from the previous cases, recall that $A$ is unramified over $R$ if and only if $A$ is formally unramified and of finite type over $R$, see Algebra, Lemma 00UU.

To prove (e), recall that by Algebra, Lemma 08WD or [EGA4, Théorème 17.6.1] the algebra $A$ is étale over $R$ if and only if $A$ is flat, unramified, and of finite presentation over $R$. □

## References

[EGA4] Grothendieck, Alexander and Dieudonné, Jean. Éléments de géométrie algébrique IV. Institute des Hautes Études Scientifiques., Publications Mathématiques, vol. 20, 24, 28, 32, 1964-1967.